School of Computer Science Engineering and Information Systems

Agile Development Process and Devops Lab

Assessment -5

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Course code: ISWE406P

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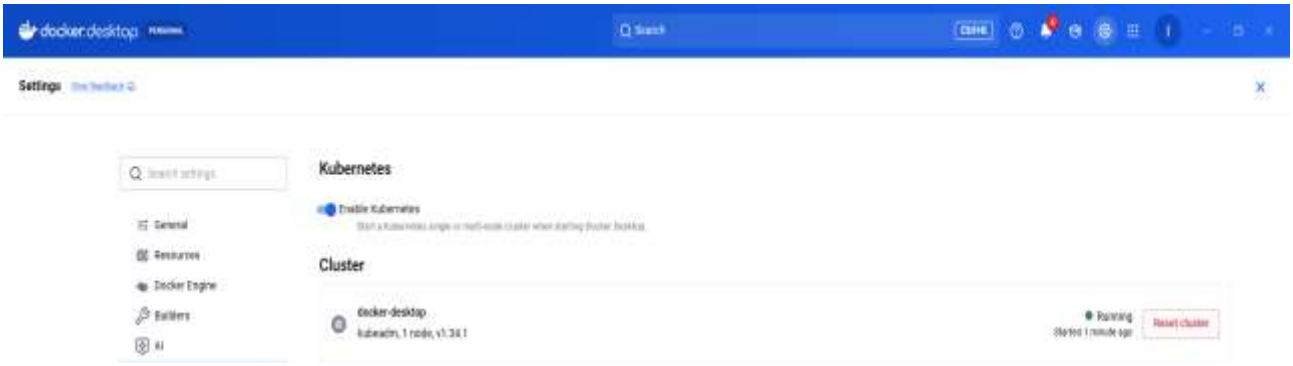
Jenkin – Docker – Kubernetes

Step1

Install Docker Desktop

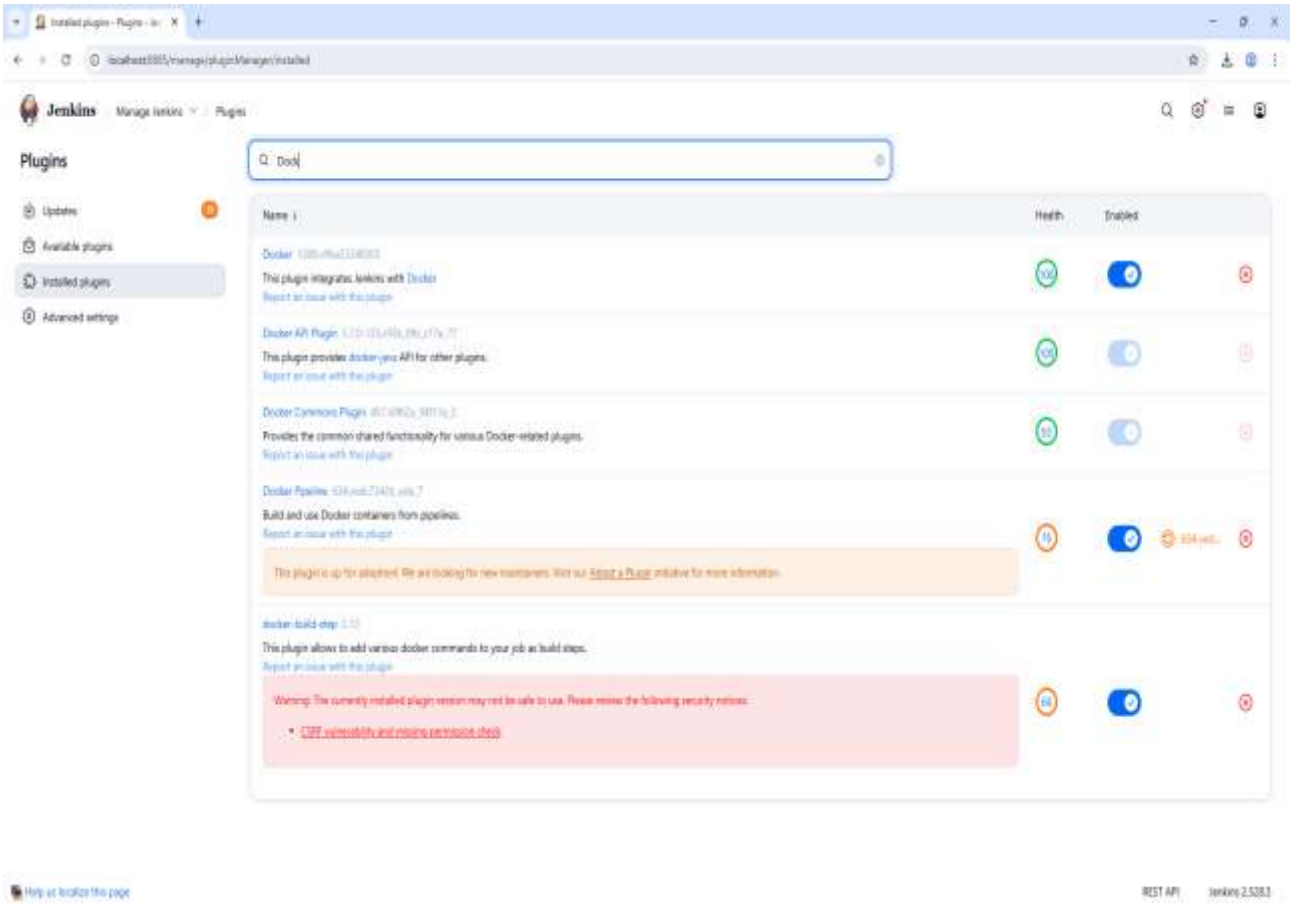
Step2

1. Enable Kubernetes
2. Open the Docker Dashboard: Launch the Docker Desktop application on your system.
3. Navigate to Settings: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.
4. Select the Kubernetes Tab: In the Settings menu, find and select the Kubernetes tab.
5. Enable Kubernetes: Check the box labeled Enable Kubernetes.
6. Apply & Restart: Click the Apply & Restart button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time



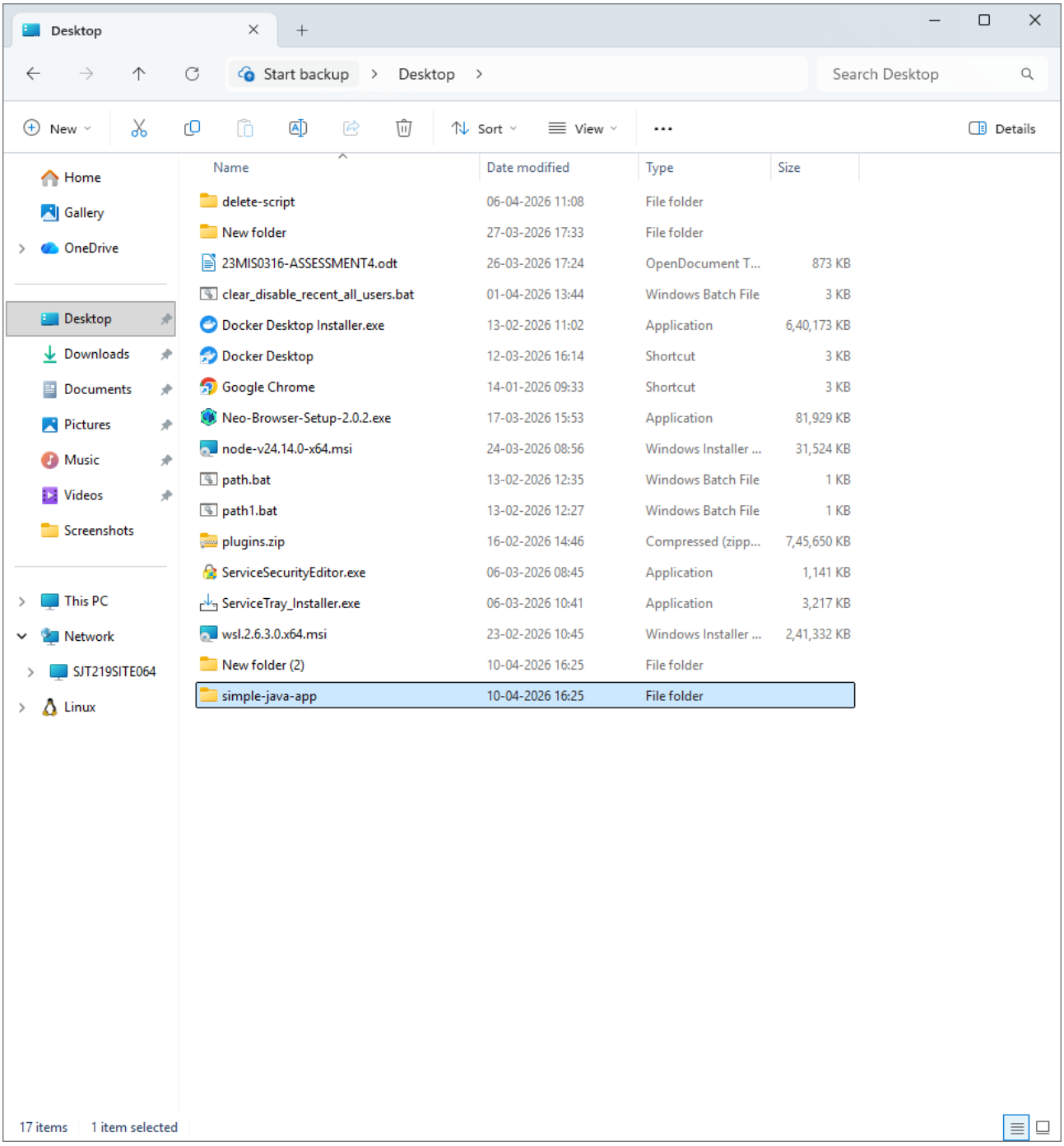
Step 3

In Jenkin, enable two plugins (Docker and Docker Pipeline) and Install them



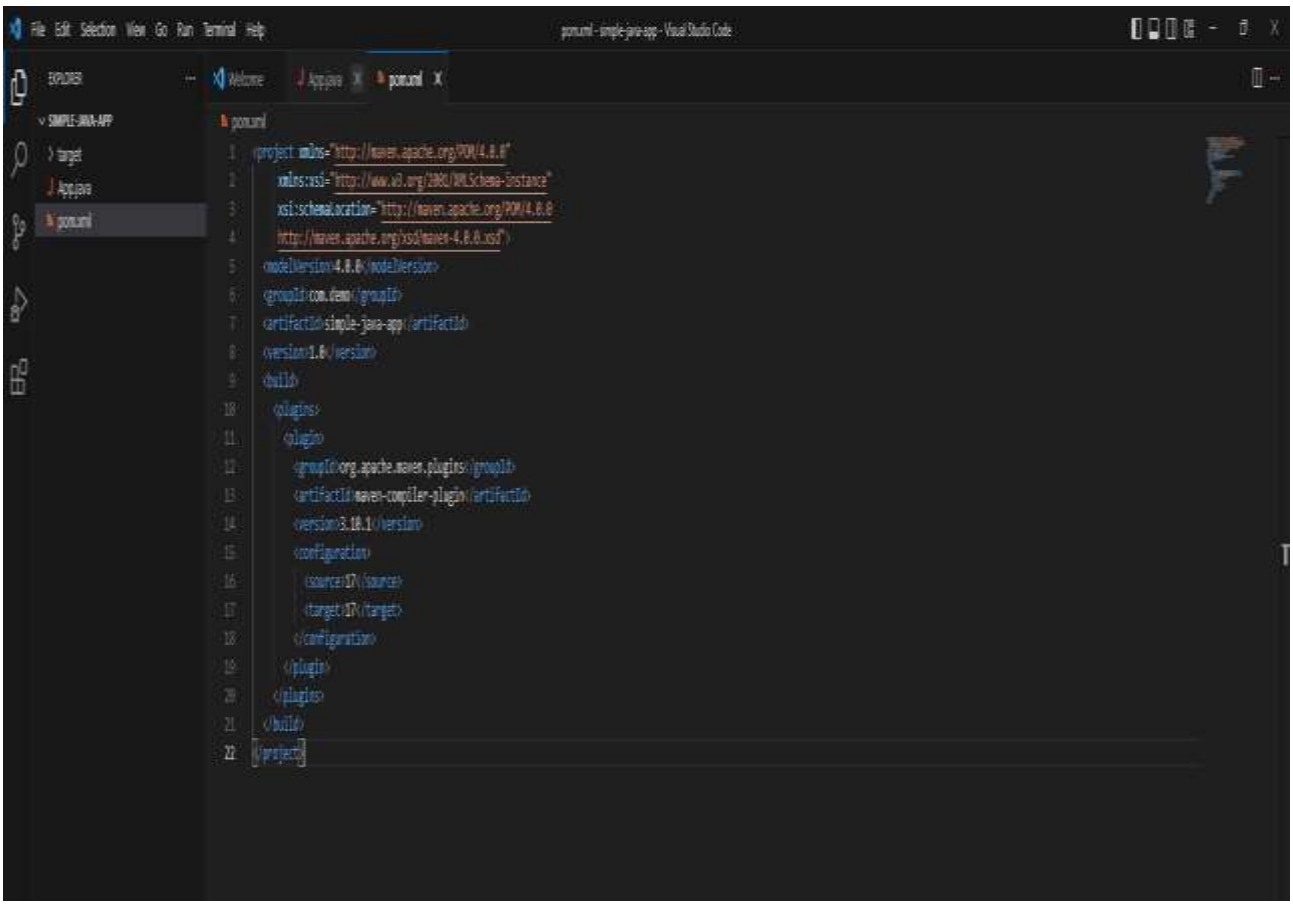
Step 4 – Create Simple Java Application

Create a folder in local machine: simple-java-app



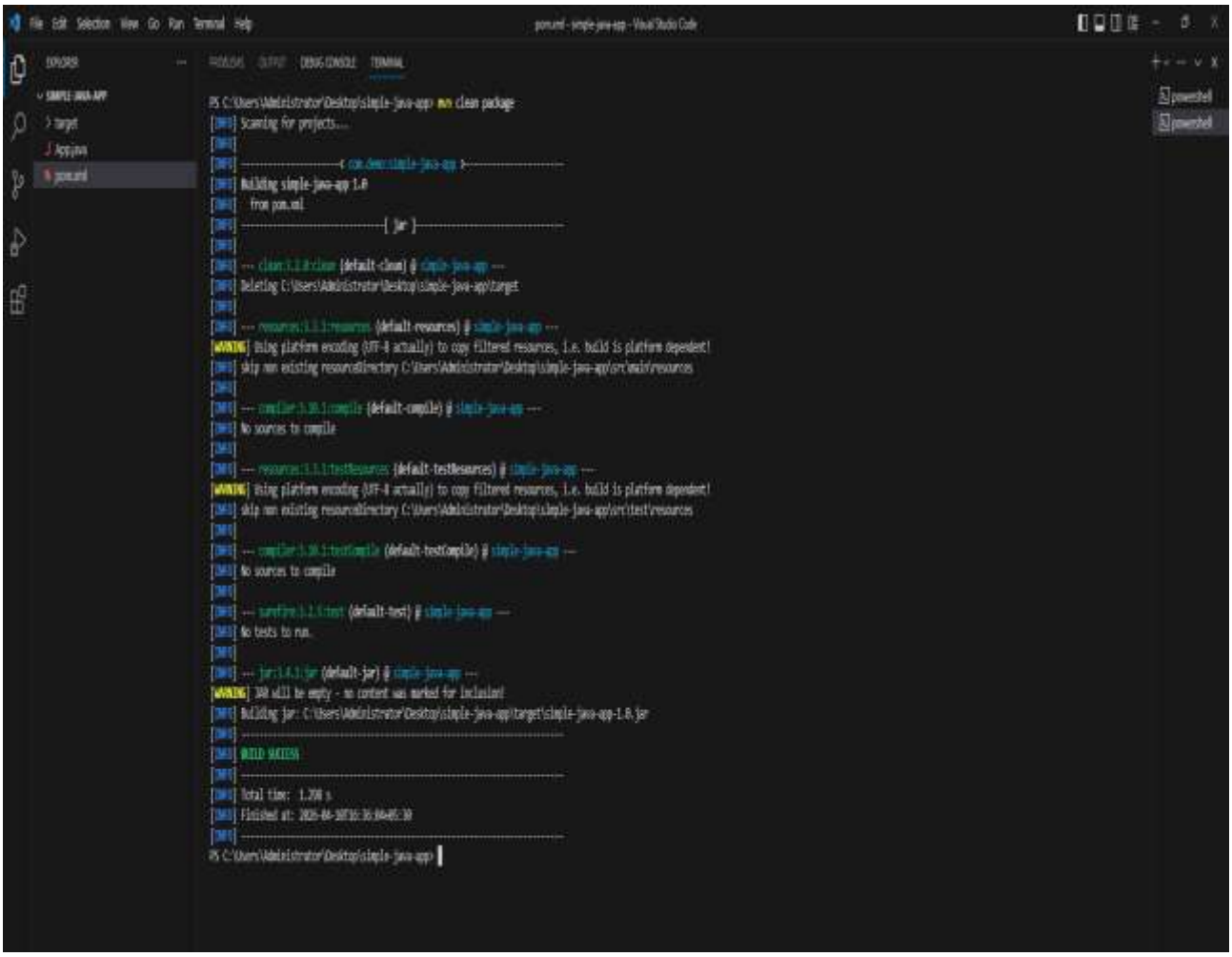
Create Java File App.java

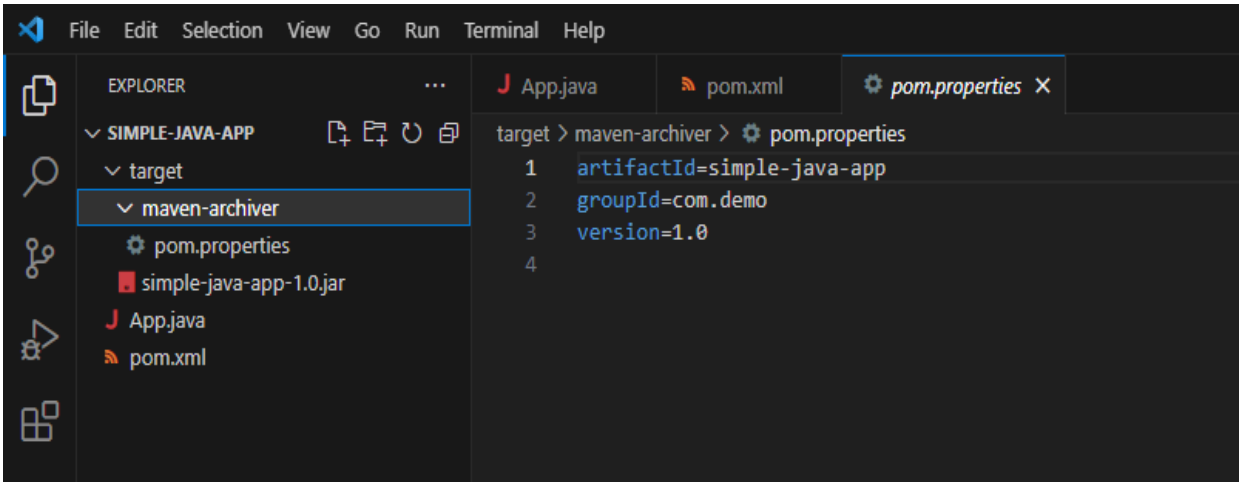
Create Maven File pom.xml



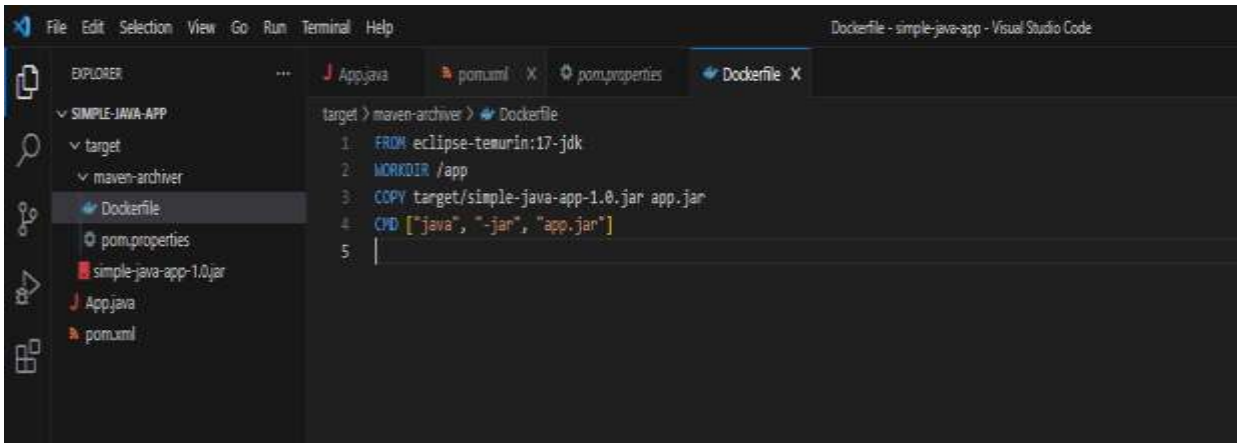
<u>Step 5</u> - Build Locally (Test) Inside project folder:

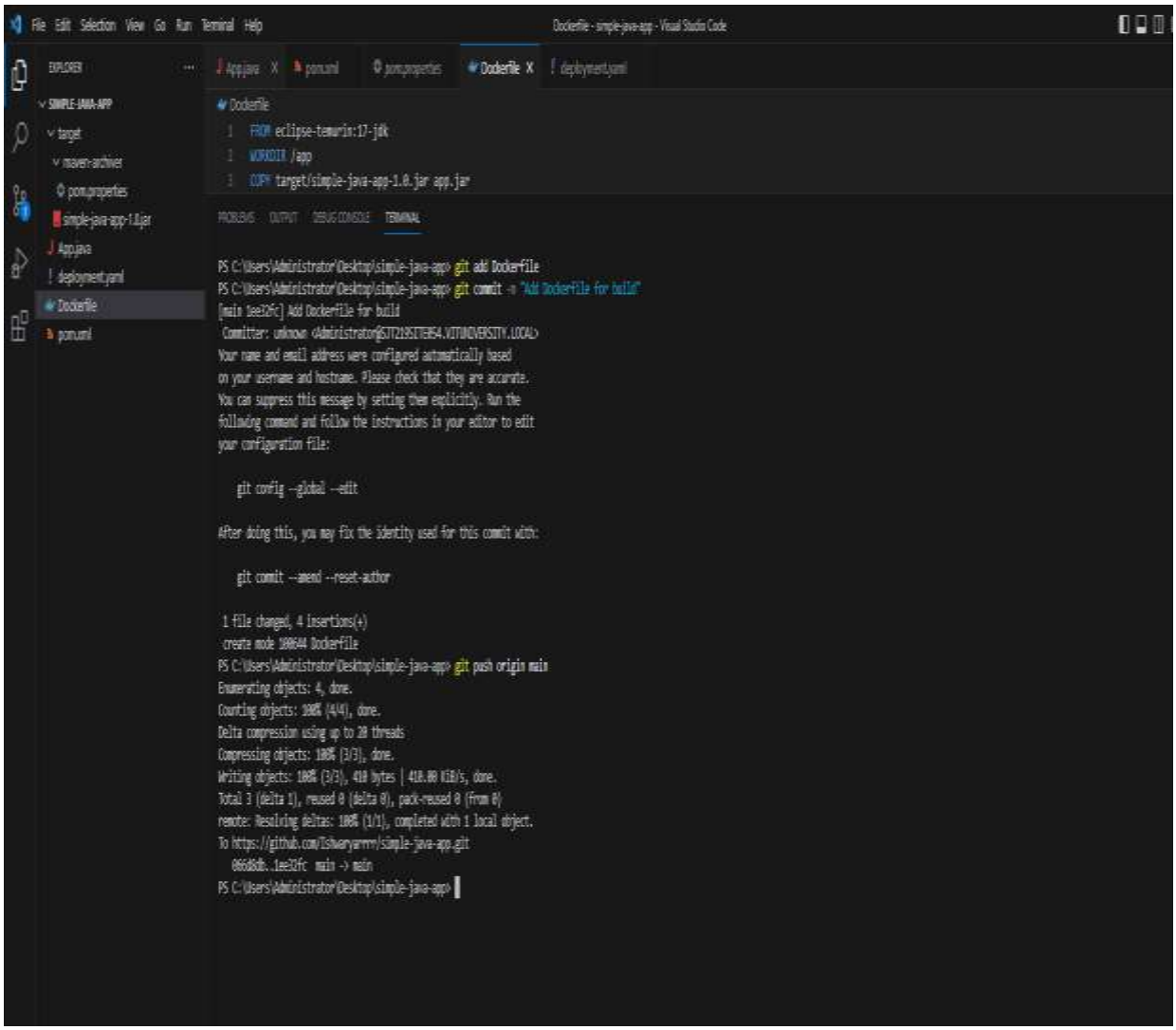
mvn clean package





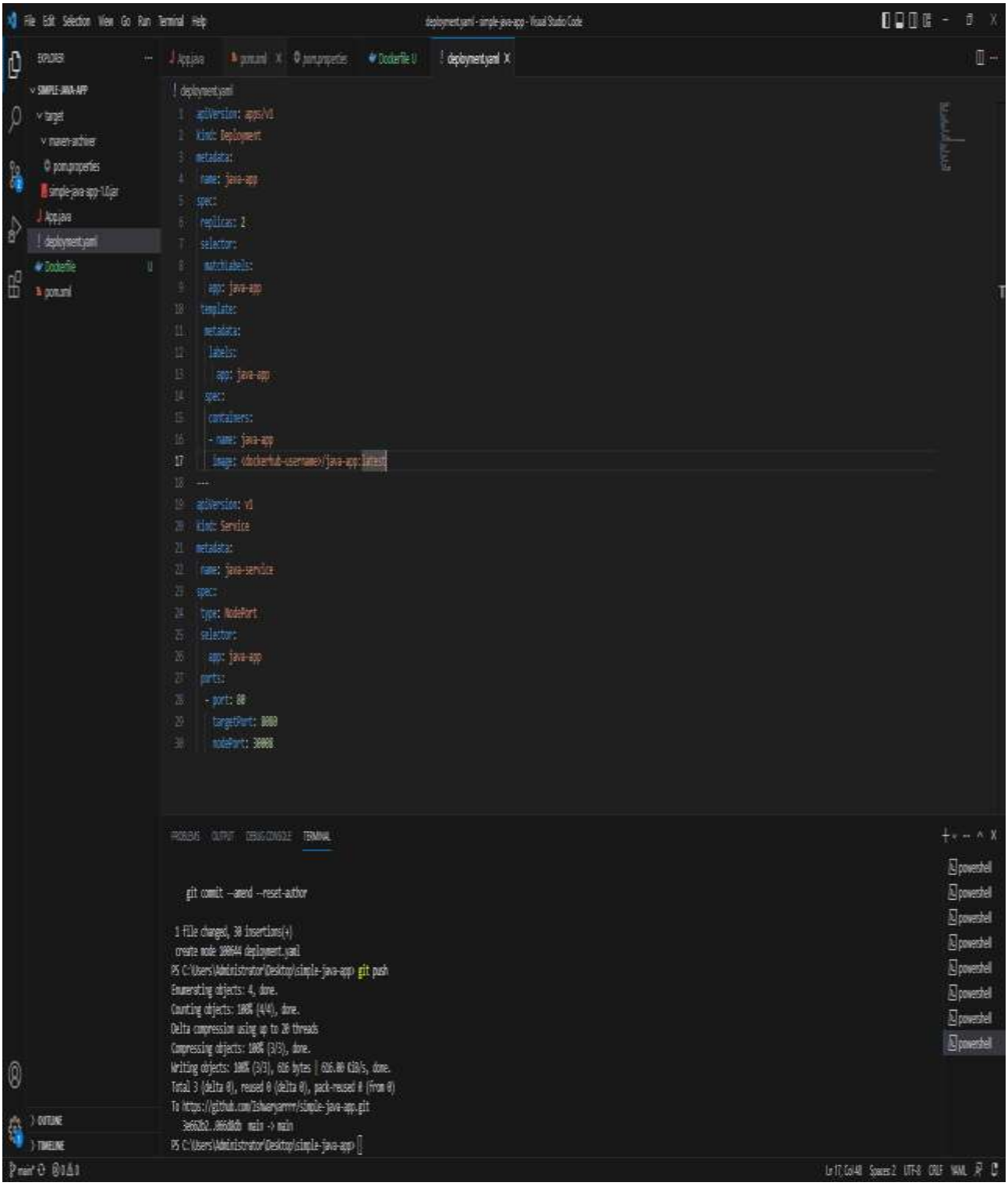
Step 6 – Create Dockerfile





Step 7 - Push Code to GitHub (Before that create a public repository in the Github)

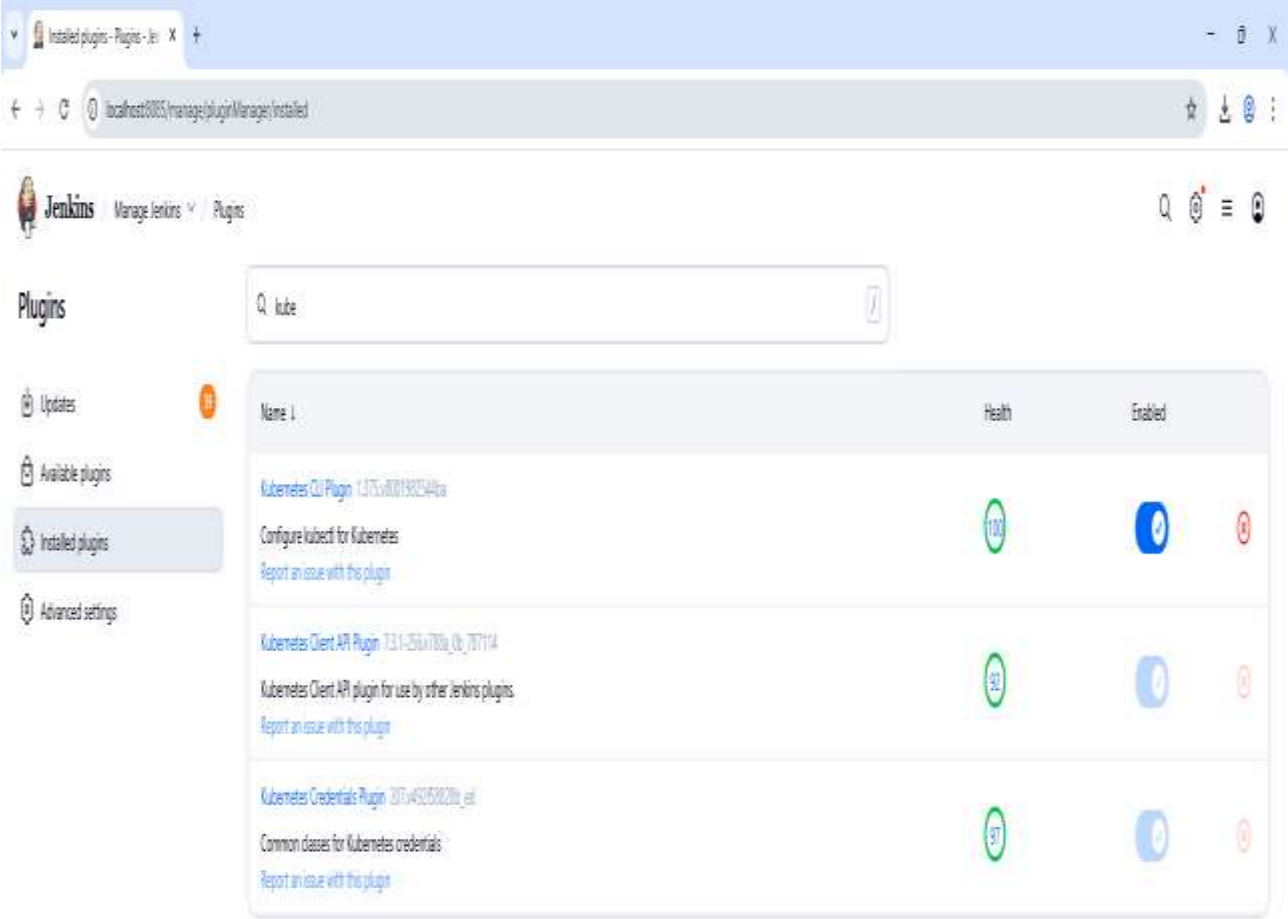
Step 8 - Create Kubernetes Deployment File



Step 9 - Configure Jenkins

Run Jenkins

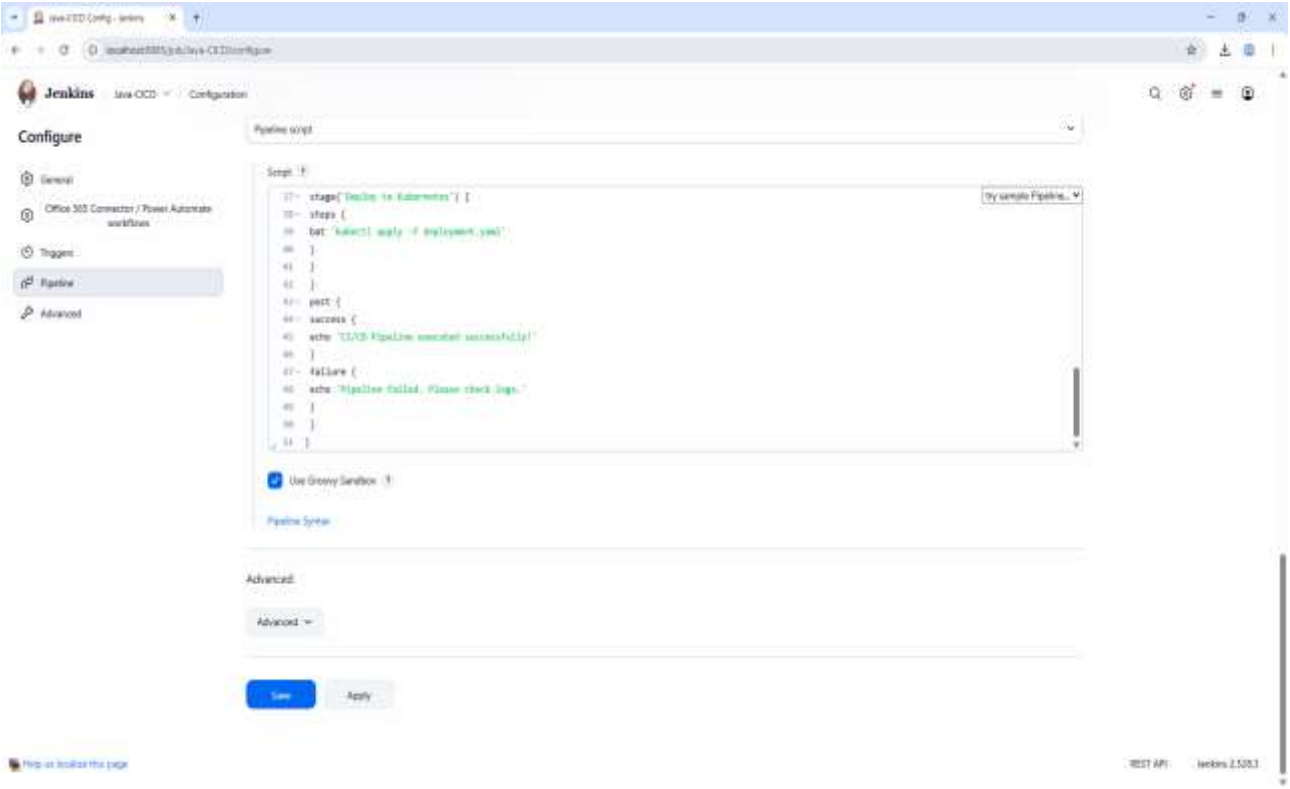
Install Kubernetes CLI plugin



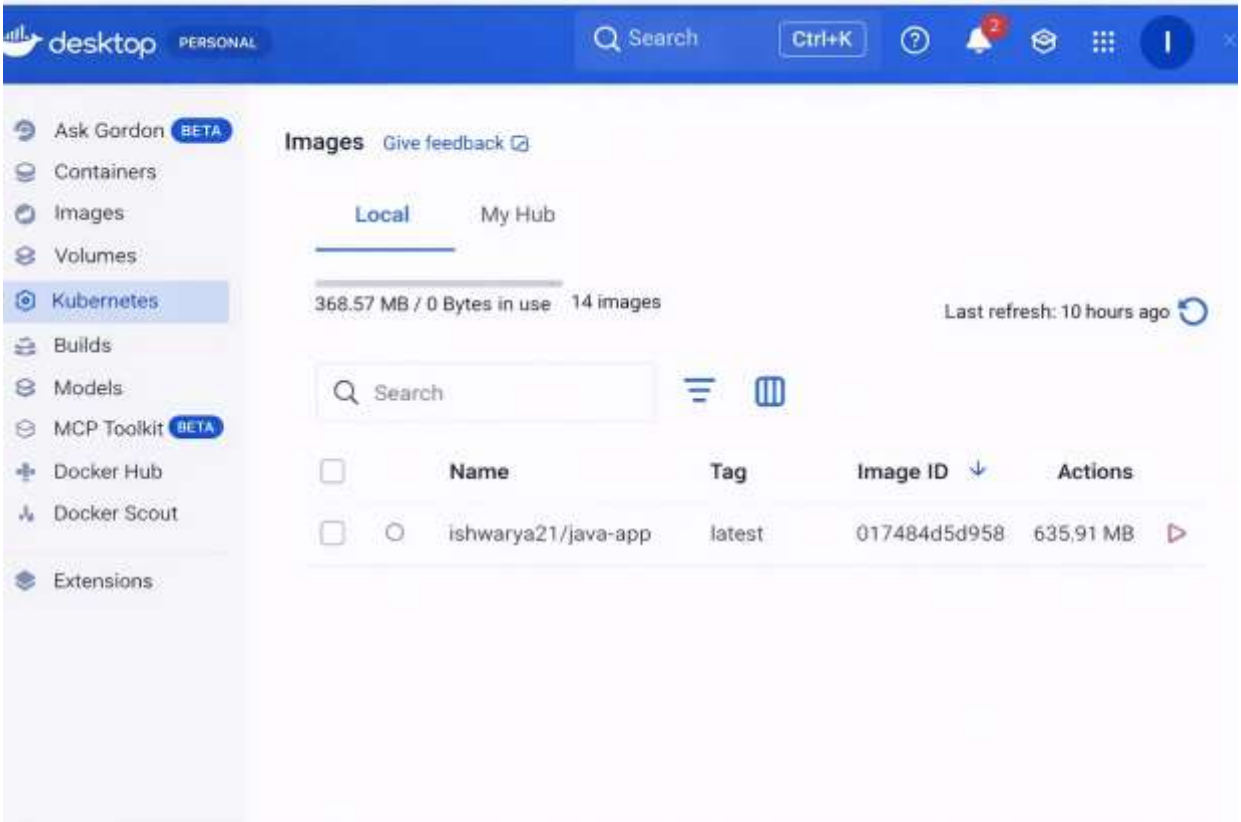
Step 10 - Add DockerHub Credentials in Jenkins

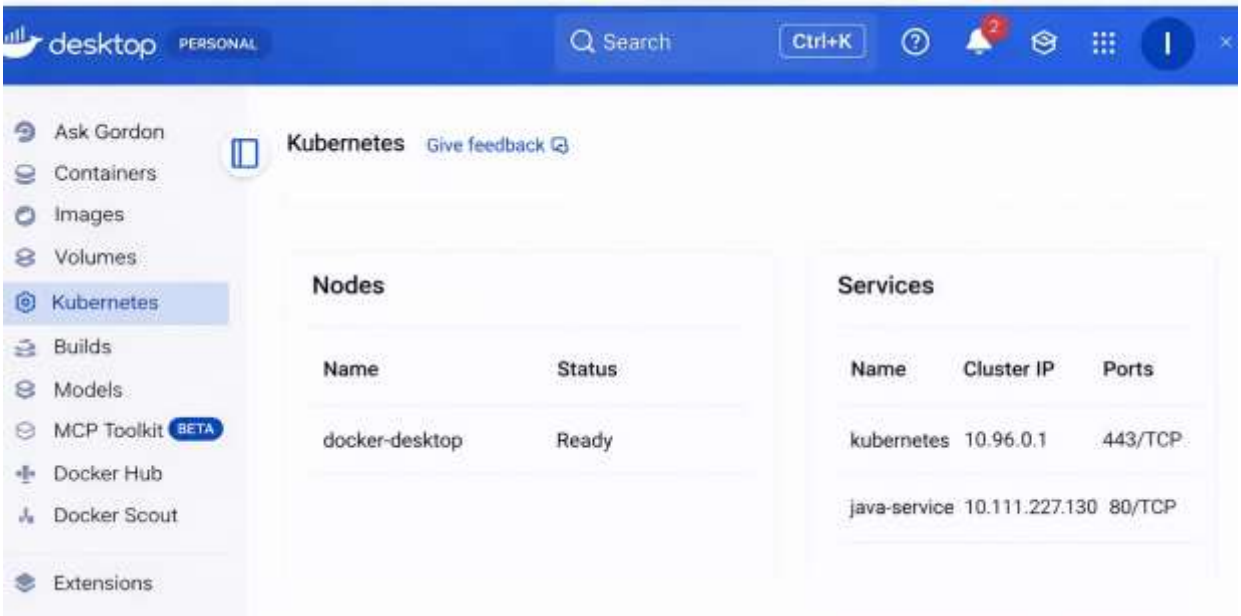
Step 11 - Create Jenkins Pipeline Job

New Item → Pipeline → Name: Java-CICD



Step 12 – Verify Docker and Kubernetes Deployment





```
PS C:\Users\Administrator\Desktop\simple-java-app> kubectl get svc
NAME                 TYPE           CLUSTER-IP     EXTERNAL-IP        AGE
java-service         NodePort     10.111.227.130   <none>           5m28s
kubernetes           ClusterIP    10.96.0.1        <none>           169m
```

Spring Boot CI/CD Pipeline (GitHub + Jenkins + Docker + Kubernetes)

Pom.xml

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>
    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.2.5</version>
        <relativePath/>
    </parent>

    <groupId>com.example</groupId>
    <artifactId>springbootproject</artifactId>
    <version>0.0.1-SNAPSHOT</version>

    <properties>
        <java.version>17</java.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
    </build>

</project>
```

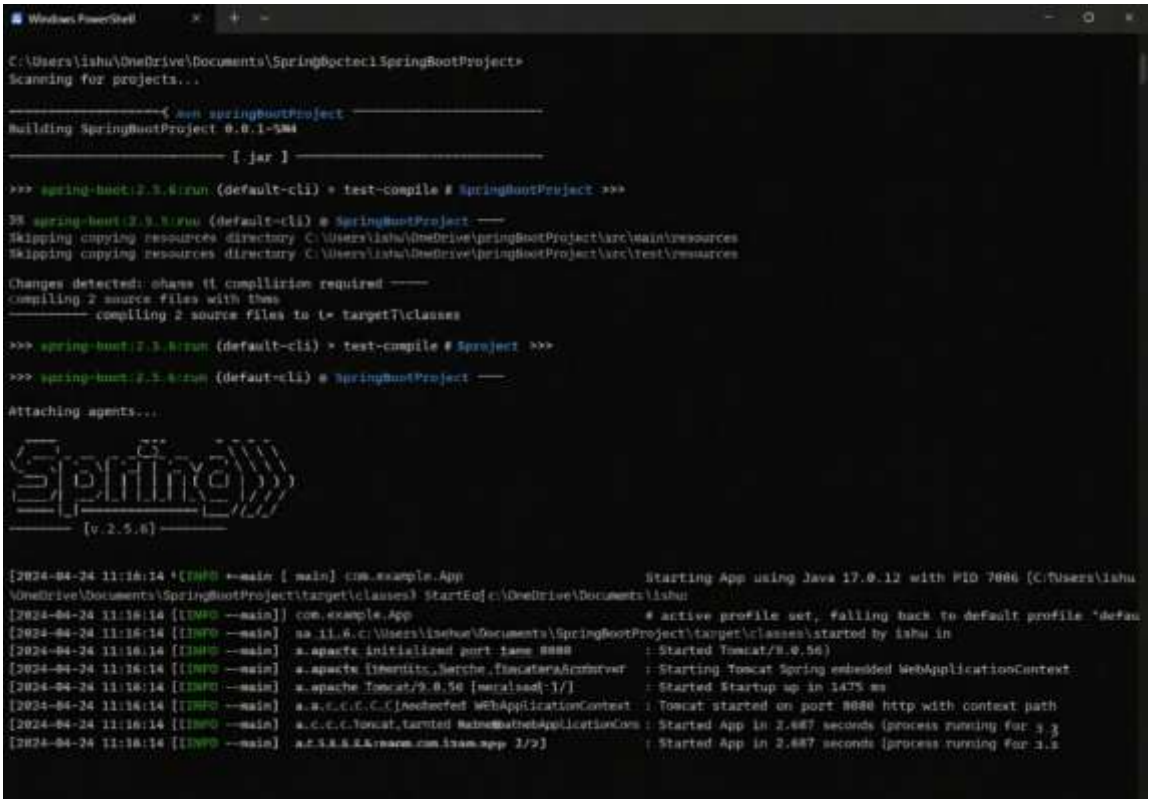
App.java

```java
package com.example;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class App {

    public static void main(String[] args) {
        SpringApplication.run(App.class, args);
    }

}
```

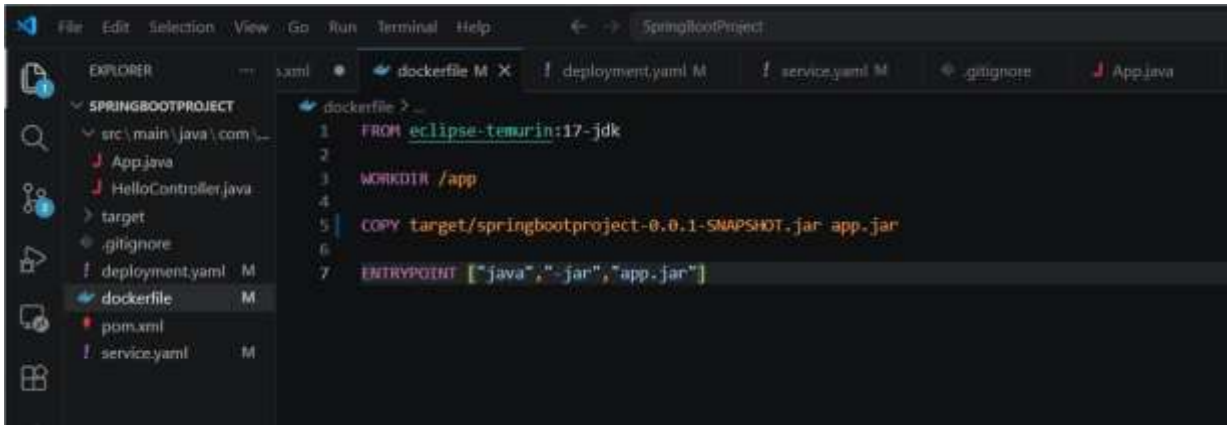
HelloController.java

```java
package com.example;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    @GetMapping("/")
    public String home() {
        return "App Running";
    }
}
```

mvn spring-boot:run



Dockerfile



```
C:\Users\ishu\OneDrive\Documents\SpringBootProject> mvn clean package
Scanning for projects...

-------------------< mvn clean package: ------------------------
Building springbootproject 0.0.1-SNAPSHOT

---------------------------[ jar ]----------------------------

[INFO] -- clean:3.0.0:clean (default-cli) @ springbootproject ---
Deleting C:\Users\ishu\OneDrive\Documents\SpringBootProject\targetaget

[INFO] -- resources:3.0.2:resources (default-resources) @ springbootproject ---
Skipping copyingsing resourceDirectory C:\Users\ishu\OneDrive\DocumentsProject\E\main\resources
Skipping copyingsing resourceDirectory C:\Users\ishu\OneDrive\DocumentsProject\E\main\resources

[INFO] -- compiler:3.1.1:compile (default-compile) @ springbootproject ---
Changes detected - recompiling the module! (Iomplimgte 15I)
Compiling 2 source files with javac [debug release 17] to target\classes

[INFO] -- compiler:3.1.1:testCompile (default-testCompile) @ springbootproject ---
Changes detected - recompiling module! (Iompiling 15I% to target\classe!
Compiling 2 source files with javac [debug release 17] to target\test-classes

[INFO] -- surefire:3.2.1:test (default-test) @ springbootproject ---
Tests run 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.146 s - in

[INFO] -- jar:3.3.0:jar (default-jar) @ springbootproject ---
Building jar: C:\Users\ishu\OneDrive\Documents\SpringBootProjecttlect\target\springbootproject

[INFO] -- spring-boot:2.5.6:repackage (default) @ springbootproject ---
Replacing main artifact with repackaged archive
Replacing C:\Users\ishu\OneDrive\Documents\SpringBootSpringBootProject\target\springbootproject-0.0.1-SNAPSHOT.jar.original

[INFO] The original artifact has been renamed to C:\Users\ishu\OneDrive\Documents\SpringBootProject-0.0.1-SNAPSHOT.jar.orignal
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
[INFO] Total time: 7.433 s
[INFO] Finished at: 2024-64-24T28-48:05+06:30
------------------------------------------------------------------------

C:\Users\ishu\OneDrive\Documents\SpringBootProject>
```

docker build -t springboot-app .

```
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> docker build -t springboot-app .
[+] Building 11.1s (9/9) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                  0.1s
 => => transferring dockerfile: 1.02kB                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                            2.4s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                               0.0s
 => [internal] laad .dockerignore                                                                     0.0s
 => => ransfnrring context: 2B                                                                        0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:2f18691aaacb80dc64b290b0257fa882576e45362ec9fb351cca72b53c999811f3  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:2f18691aaacb80dc4b29981b270fa882576e45362ec9fb351cca72b53c999811f3  0.1s
 => [internal] load build context                                                                     0.0s
 => => transferring context: 39.77kB                                                                  0.0s
 => CACHED [2/3] WONKDIR /app                                                                         0.0s
 => [3/3] COPY target/springbootproject-0.0.1-SNAPSHOT.jar app.jar                                    0.1s
 => exporting to image                                                                                3.1s
 => => exporting layers                                                                               3.0s
 => => exporting manifest sha256:1259e1125ef1ff4f8ab417bfb3960d8dac42ee5c9a9e1b8689b4b51a4f67fdeb4ab  0.0s
 => => exporting config sha256:9d0fib71I0b5d45ec5b6a8108d113be34aaa407fe31e3b440d1867bc86e8016cbeab  0.0s
 => => exporting attesstaing attestation manifest sha256:bf7454fd12157aaaa14e88ec17ea3b56bdffe72fb67bb889e58a5f7d613ea  0.0s
 => => exporting manifest list sha256:8d2a6:802b07ef832670bf4fbba31192b09af778aeic4c808bdfd39826c8b8ccbcl1a8  0.0s
 => => naming to danifeer iol/library/springboot-app:latest                                           0.0s
 => => unpacking to docker.io/library/springbooot-app:latest                                          0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/halrev3de0ntei0e0llarzuofq
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject>
```

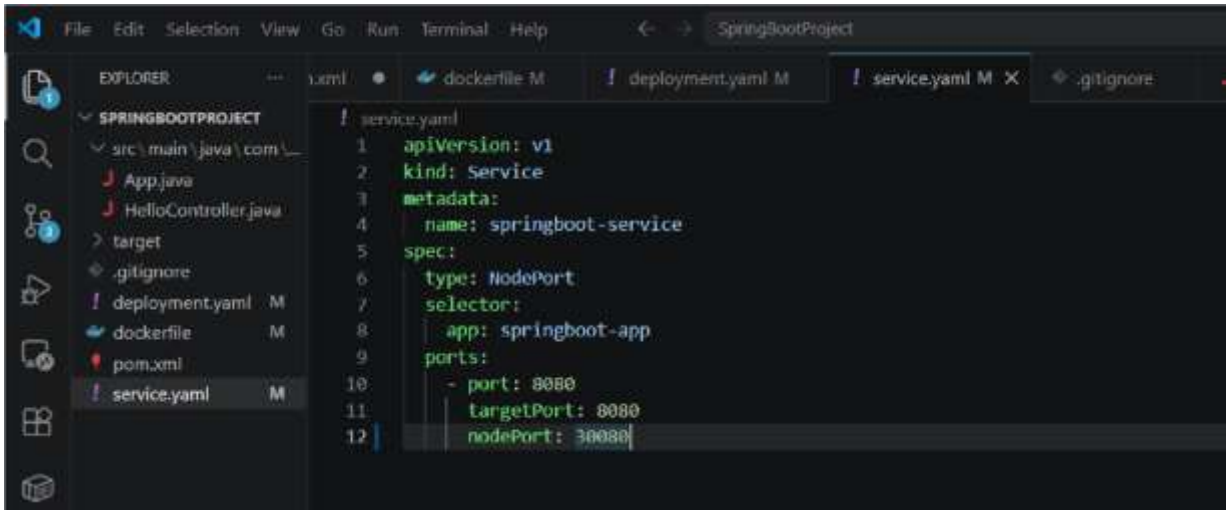
docker run -p 8081:8080 springboot-app

```
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> docker run -p 8080:8080 springboot-app

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v2.5.6)

2024-04-24T20:56:14.542+05:30  INFO 1 ---        ! Starting App v0.0.1-SNAPSHOT using Java 17.0.10 with PID 1
(/app/springbootproject-0.0.1-SNAPSHOT.jar started by root in /app)

2024-04-24T20:56:14.550+05:30  INFO 1 ---        ! Starting profile set, falling back to 1 default profile:"default"
2024-04-24T20:56:17.464+05:30  INFO 1 --- [1 min ] com.example.App

2024-04-24T20:56:17.464+05:30  INFO 1 --- [1 main] com.example.App  : Starting App v0.0.1-SNAPSHOT using Java
17.0.10 with PID 1
2024-04-24T20:56:17.464+05:30  INFO 1 --- [1 main] com.example.App  : No active profile set, faling back to 1
    default profile: "default"
2024-04-24T20:56:17.464+05:30  INFO 1 --- [1 main] o.s.b.w.embedded.tomcat.TomcatWebServer
2024-04-24T20:56:17.468+05:30  INFO 1 --- [1 main] org.apache.catalina.core.StandardService
2024-04-24T20:56:17.470+05:30  INFO 1 --- [1 main] org.apache.catalina.core.StandardEngine
                                                  : Starting Servlet engine: [Apache Tomcat/9.0.20]
2024-04-24T20:56:17.531+05:30  INFO 1 --- [1 main] o.s.b.w.s.ServletHebServerApplicationContext : Inialization
2024-04-24T20:56:17.623+05:30  INFO 1 --- [1 main] o.s.b.w.embedded.tomcat.TomcatWebServer
2024-04-24T20:56:17.623+05:30  INFO 1 --- [1 main] com.example.App  : Started App in 2.513 seconds
                                                   (process running for 2.978)
2024-04-24T20:56:17.625+05:30  INFO 1 --- [1 main] o.s.w.s.DispatcherServlet : Initializing Spring DispatchServlet
2024-04-24T20:56:17.628+05:30  INFO 1 --- [1 main] completed initialization in 10 ms

PS C:\Users\ishu\OneDrive\Documents\SpringBootProject>
```

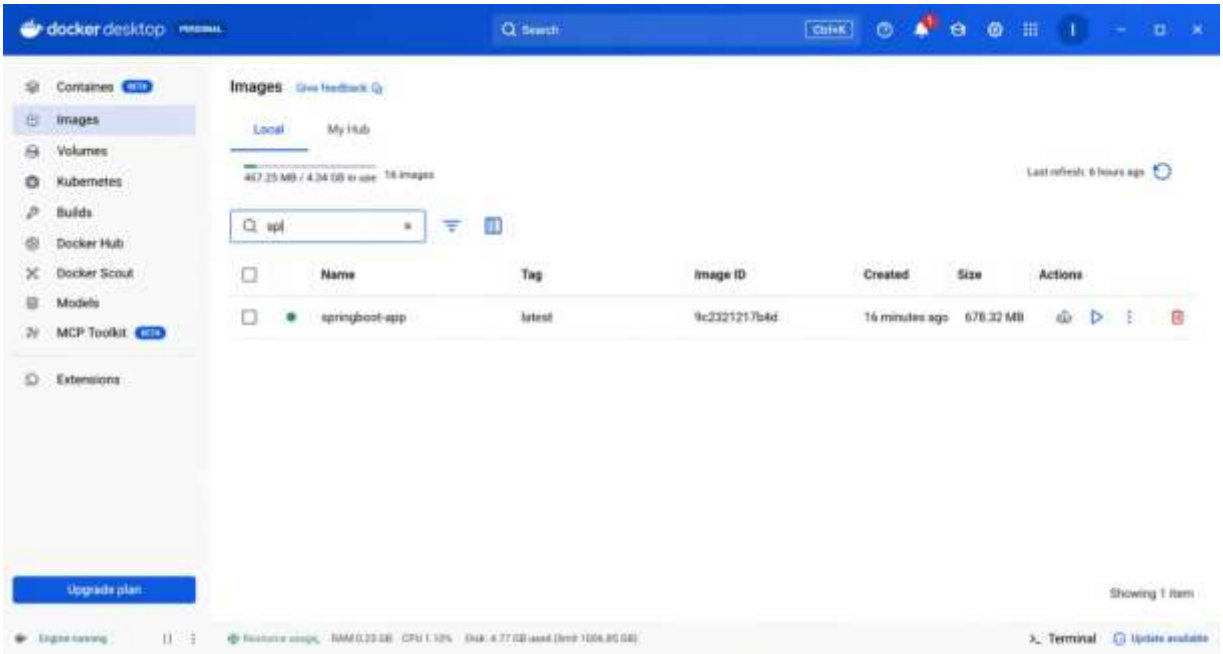
Kubernetes deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: springboot-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: springboot-app
  template:
    metadata:
      labels:
        app: springboot-app
    spec:
      containers:
        - name: springboot-app
          image: springboot-app:latest
          imagePullPolicy: Never
          ports:
            - containerPort: 8080
```

Service.yaml

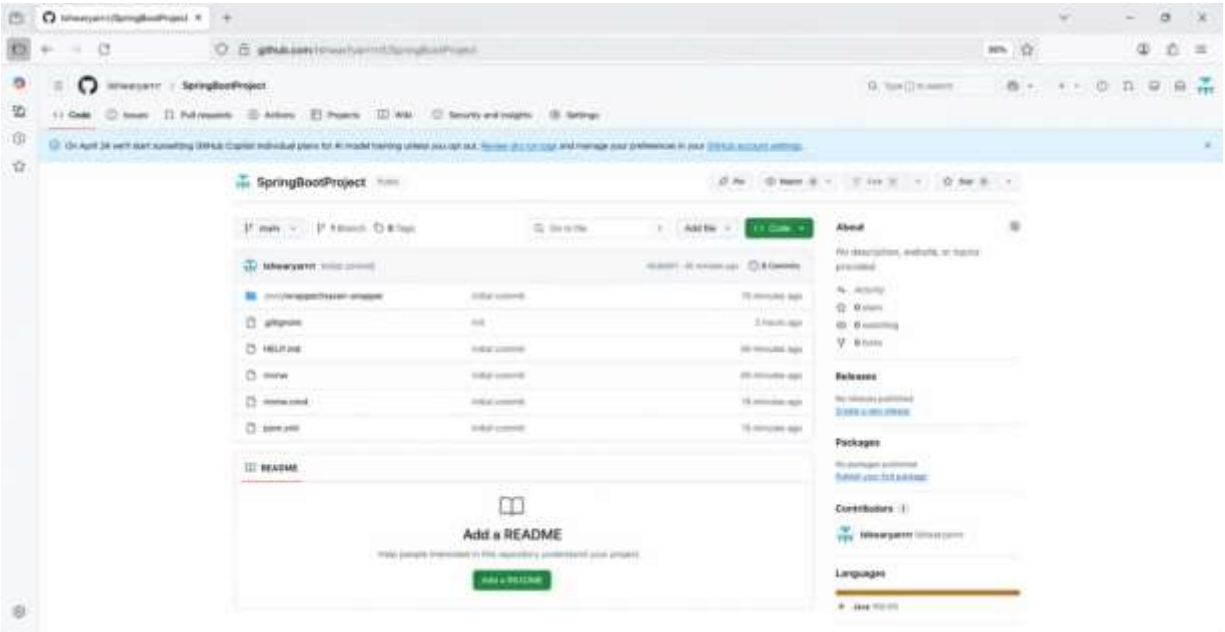



```
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> kubectl get nodes
NAME                   STATUS   ROLES           AGE   VERSION
docker-desktop         Ready    control-plane   2m45s   v1.34.1

PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> kubectl apply -f deployment.yaml --validate=
deployment.apps/springboot-app created

PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> kubectl apply -f service.yaml --validate=fa
service/springboot-service created

PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> kubectl apply -f service.yaml --validate=fa
service/springboot-service created
```

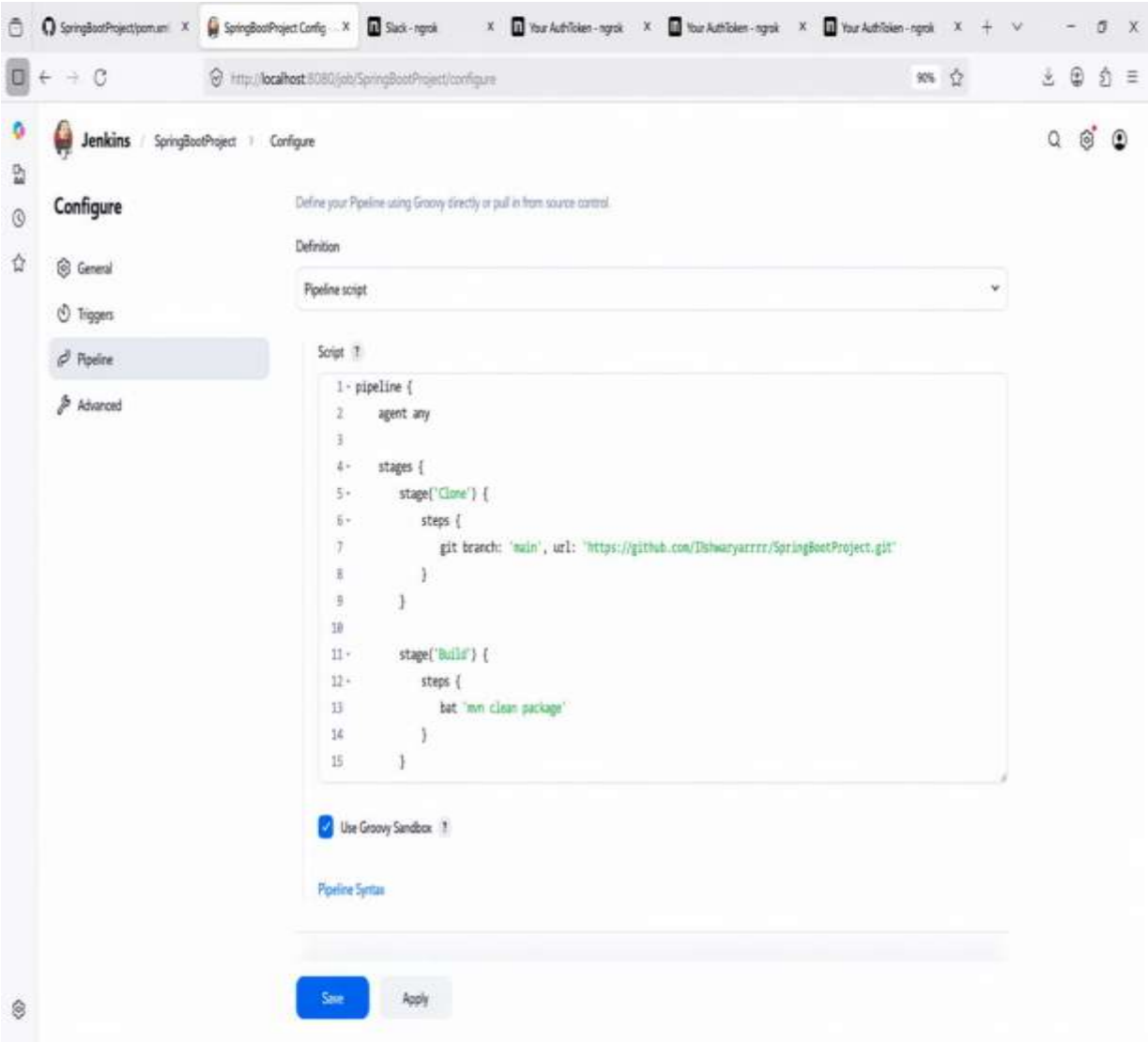


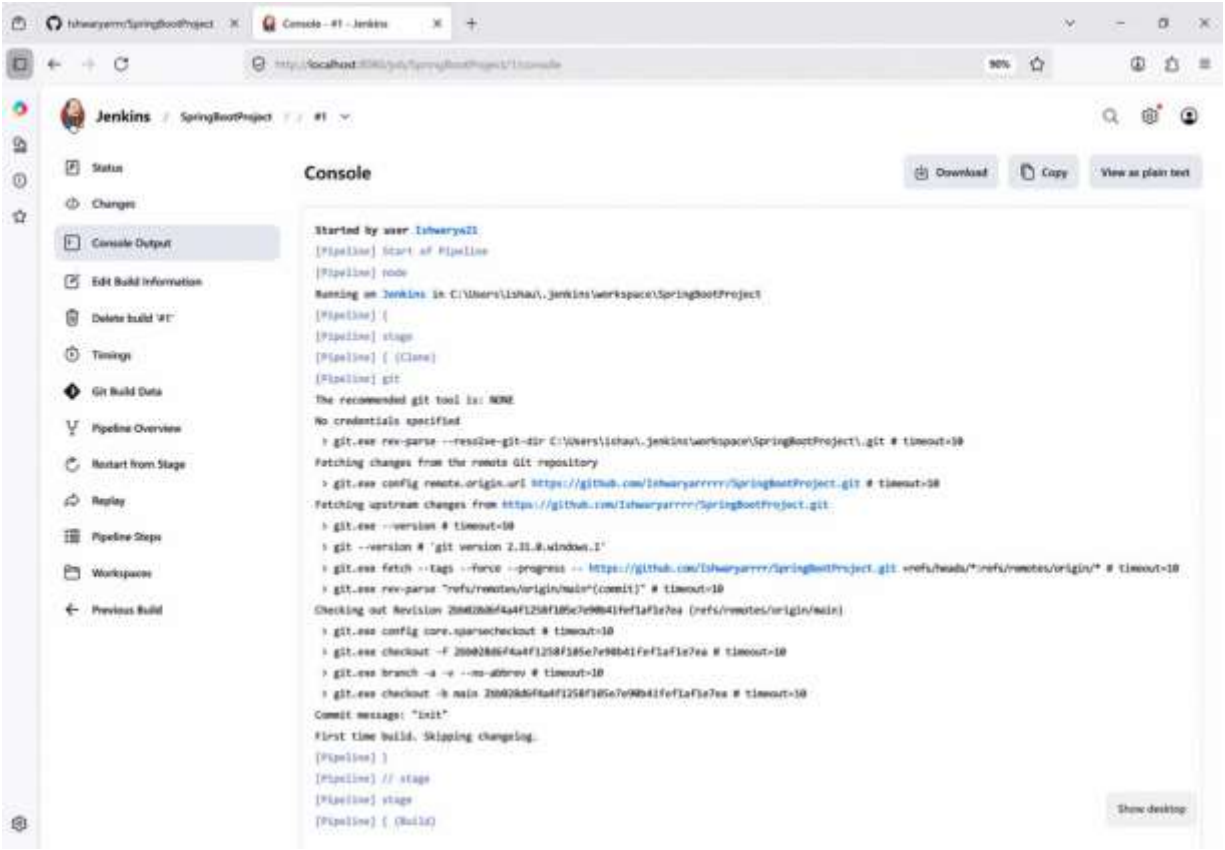
GitHub

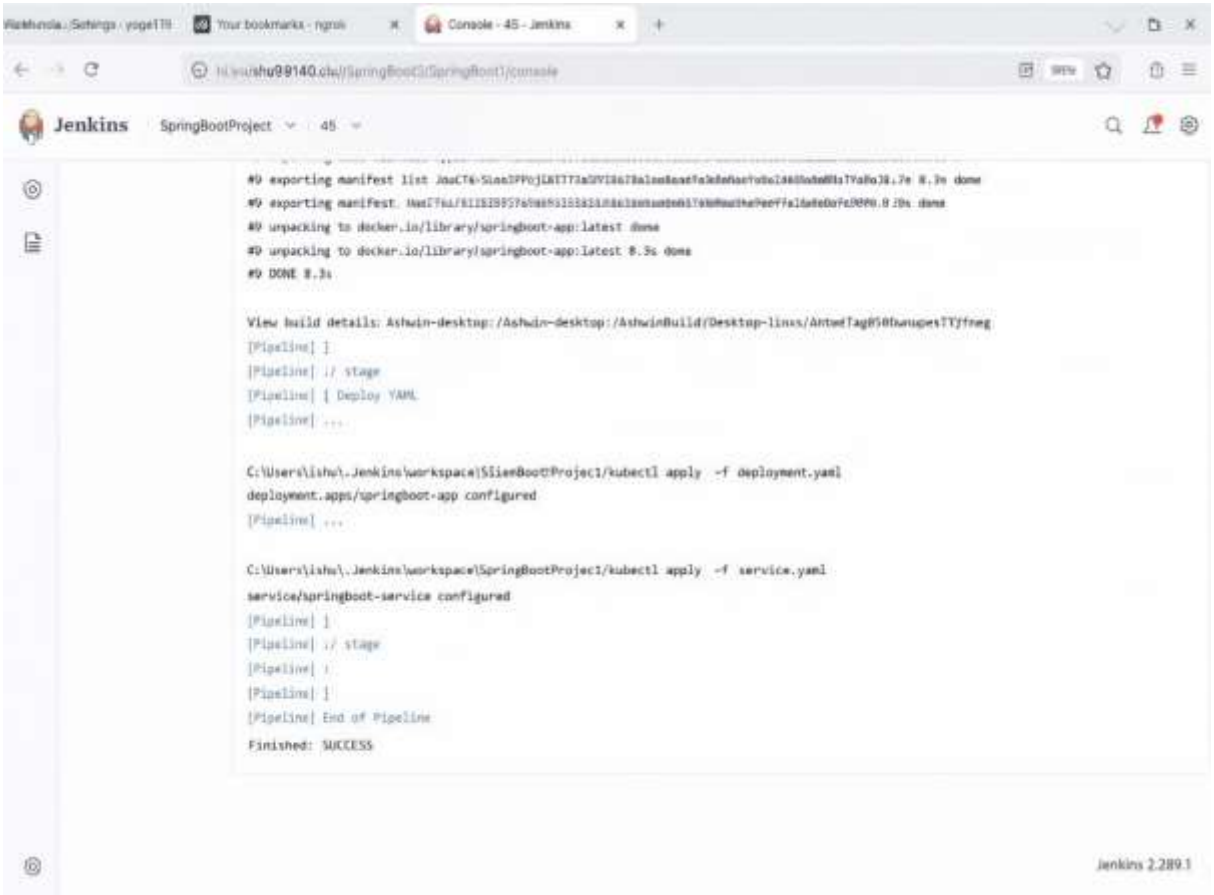
```
PS C:\Users\Ishwaryarrrr\OneDrive\Documents\SpringBootProject> git add .
PS C:\Users\Ishwaryarrrr\OneDrive\Documents\SpringBootProject> git commit -m "init"
[master 2bb028d6] init
4 files changed, 4 insertions(+), 7 deletions(-)
  create mode 100644 .gitignore
  create mode 100644 target/class/com/example/App.class
  delete mode 100644 target/classes/com/example/HelloContoller.class
  delete mode 100644 target/maven-archiver/pom.properties
  delete mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
  delete mode 100644 target/maven-status/maven-compiler-plugin/compile
  rename target/springbootproject-0.0.1-SNAPSHOT.jar.original
PS C:\Users\Ishwaryarrrr\OneDrive\Documents\SpringBootProject> git remote add https://github.com/Ishwaryarrrr-sudo/SpringBootProject.git

PS C:\Users\Ishwaryarrrr\OneDrive\Documents\SpringBootProject> git branch -M main
Enumerating objects: 32, done.
Delta compression using up to 8 threads
Writing objects: 100% (21/21), done.
Writing objects: 100% (32/32), 16.90 KiB 3.40 MiB/s, done.
Total 32 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Ishwaryarrrr-sudo/SpringBootProject.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Ishwaryarrrr\OneDrive\Documents\SpringBootProject>
PS C:\Users\Ishwaryarrrr\OneDrive\Documents\SpringBootProject>
```



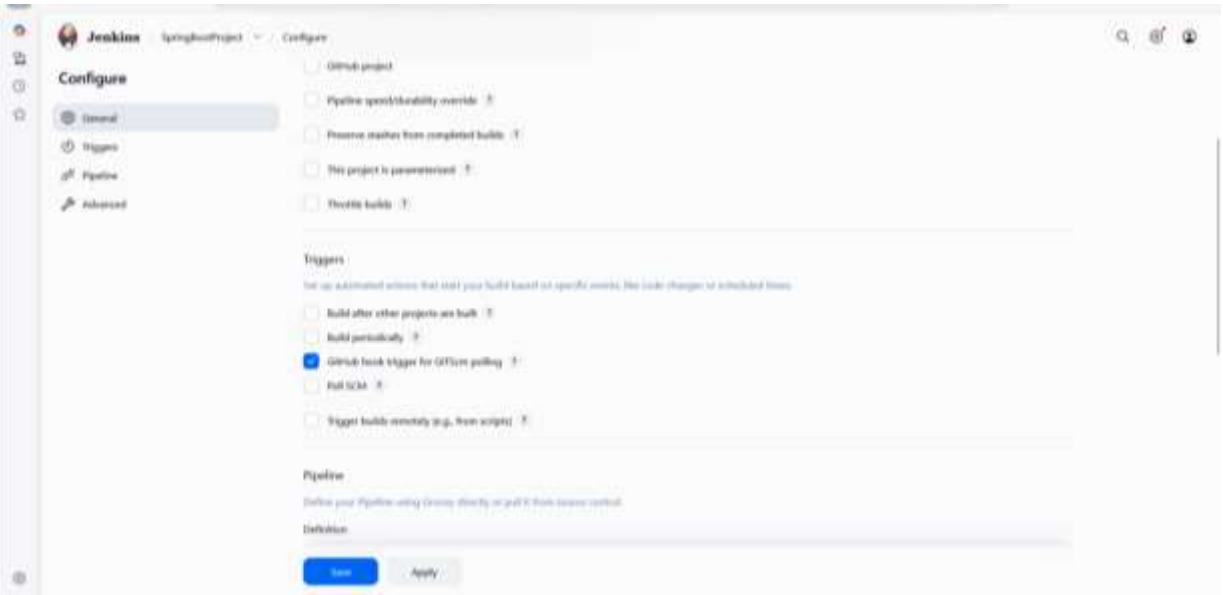
Jenkins Pipeline



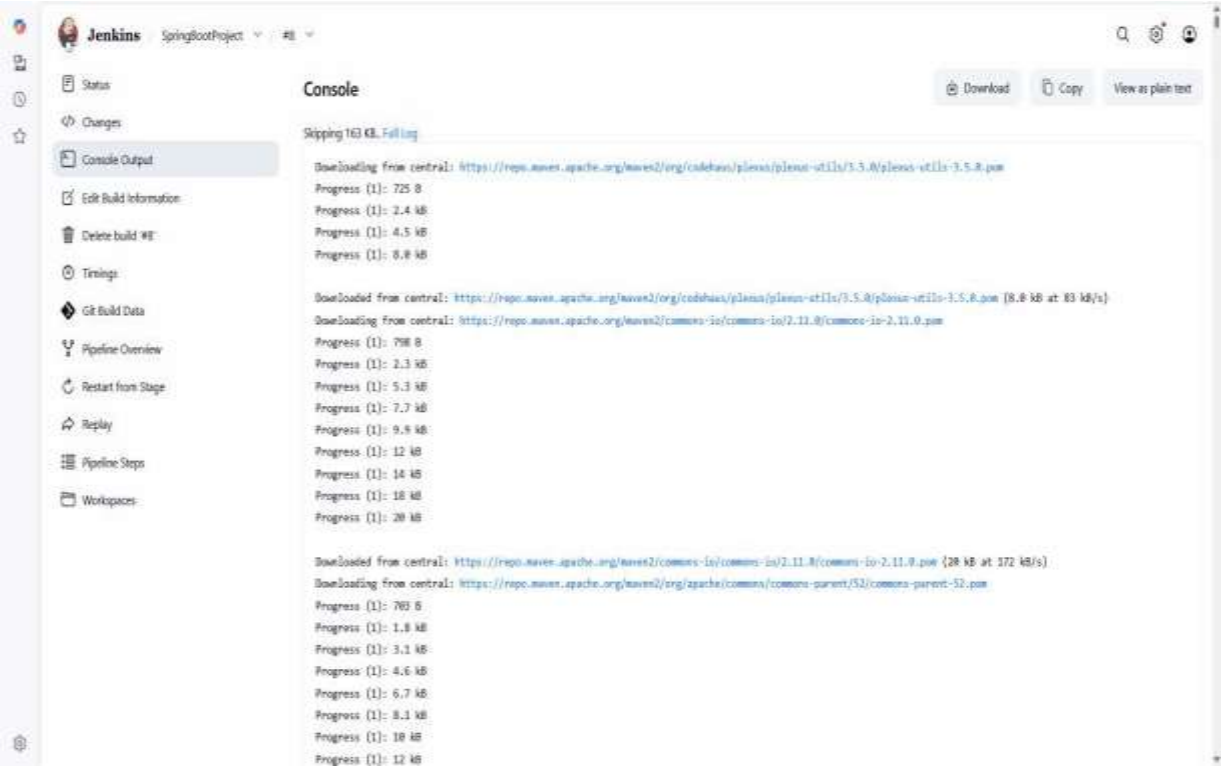

Console - #1 - Jenkins
Jenkins / SpringBootProject / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspace
Previous Build
Console
Download
Copy
View as plain text
Started by user Ishwarya21
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\ishu\.jenkins\workspace\SpringBootProject
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\Users\ishu\.jenkins\workspace\SpringBootProject\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/Ishwaryarrrrr/SpringBootProject.git # timeout=10
Fetching upstream changes from https://github.com/Ishwaryarrrrr/SpringBootProject.git
> git.exe --version # timeout=10
> git --version # 'git version 2.31.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/Ishwaryarrrrr/SpringBootProject.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 20002b86f4a4f1258f185e7e90b41fef1af1e7ea (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 20002b86f4a4f1258f185e7e90b41fef1af1e7ea # timeout=10
> git.exe branch -a -v --no-abbrev # timeout=10
> git.exe checkout -b main 20002b86f4a4f1258f185e7e90b41fef1af1e7ea # timeout=10
Commit message: "init"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
Show desktop


Your bookmarks - ngrok
Console - 45 - Jenkins
Jenkins
SpringBootProject
45
#9 exporting manifest list
#9 exporting manifest
#9 unpacking to docker.io/library/springboot-app:latest done
#9 unpacking to docker.io/library/springboot-app:latest 8.3s done
#9 DONE 8.3s
View build details: Ashwin-desktop:/Ashwin-desktop:/AshwinBuild/Desktop-linux/
[Pipeline] }
[Pipeline] // stage
[Pipeline] { Deploy YAML
[Pipeline] ...
C:\Users\ishu\.Jenkins\workspace\SpringBootProject1/kubectl apply -f deployment.yaml
deployment.apps/springboot-app configured
[Pipeline] ...
C:\Users\ishu\.Jenkins\workspace\SpringBootProject1/kubectl apply -f service.yaml
service/springboot-service configured
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] }
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.289.1

Webhook



```
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> git add .
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> git commit -m "webhook test"
[main 0686613] webhook test
 3 files changed, 7 insertions(+), 8 deletions(-)
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject> git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 664 bytes | 110.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/yoga979144-sudo/SpringBootProject.git
   2bb02b6..0686613  main -> main
PS C:\Users\ishu\OneDrive\Documents\SpringBootProject>
```



ishu91144.ncl.cu/SpringBoot1 × Console - 45 - Jenkins × +

http://ishu91144.ncl.cu/SpringBoot1/console 80%

Jenkins SpringBootProject 45

```
## exporting nonfast link decfS3.ca5cDTC37C3bhFaI71F1swN0ak80ND0C8Feh01e#474à1a1et=1:done"
## exporting manifest sha256:b36c03777o7c4b0317b0881b156881e13a834090ee3b070e1f558886708 9-0s done
## unpacking to docker.lo/library/springboot-app:latest done
## unpacking to docker.io/library/springboot-app:latest 8.3
## unpacking to docker.io/library/springboot-app:latest 吾 8.3s done
## DONE 8.3s

View build details: Ashwin-desktop:/Ashwin-desktop:AshwinBuild/Desktop-linxs/ahwinTag65800uwanapes JTyFnungc
[Pipeline] /
[Pipeline] // stage
[Pipeline] | Deploy YAML
[Pipeline] ....

C:\Users\ishu\.Jenkins\workspace\SpringBootProject1\kubectl apply -f deployment.yaml
deployment-apps/springboot-app unchanged
[Pipeline] // stage
[Pipeline] ...

C:\Users\ishu\.Jenkins\workspace\SpringBootProject1\kubectl apply -f service.yaml
service/springboot-service unchanged
[Pipeline] // stage
[Pipeline] /...
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.289.1